Jain Heritage Schools
Preparatory Examinations-1

Science · Total Marks: **50 / 50** · 25 questions

Student Name: _____ Class: 10 Date: 2026-10-20

Academic year: 2026-27

Duration: 2 hours

Instructions:

- Attempting all questions are mandatory.
- Every Topics has their own weightage as mentioned before.

Section — 1-mark questions

Question 1 1 mark Easy Chemistry MCQ

1. What is the chemical symbol for sodium?

A. So
B. S
C. Na
D. N

Question 2 1 mark Easy Biology MCQ

2. Which organelle is commonly called the powerhouse of the cell?

A. Nucleus
B. Mitochondrion
C. Ribosome
D. Vacuole

Question 3 1 mark Easy Chemistry MCQ

3. The atomic number of an element equals the number of:

A. neutrons
B. protons
C. electron shells
D. nucleons

Question 4 1 mark Easy Chemistry MCQ

4. A solution with pH 5 is generally:

A. acidic
B. neutral
C. alkaline
D. always pure water

Question 5 1 mark Easy Chemistry MCQ

5. What is the chemical formula of water?

A. HO
B. H_2O
C. H_2O_2
D. OH_{22}

Question 6 1 mark Easy Chemistry MCQ

6. Rusting of iron generally requires oxygen and:

A. helium
B. moisture
C. pure nitrogen only
D. sugar

Question 7 1 mark Medium Physics MCQ

7. When a constant force acts parallel to displacement, work done is calculated as:

A. W = F / d
B. W = F × d
C. W = m / a
D. W = v / t

Question 8 1 mark Medium Chemistry MCQ

8. When many metals react with dilute acids, which gas is commonly released?

A. oxygen
B. nitrogen
C. hydrogen
D. chlorine

Question 9 1 mark Medium Biology MCQ

9. Which organs filter blood and help produce urine?

A. lungs
B. kidneys
C. stomach
D. pancreas

Question 10 1 mark Medium Biology MCQ

10. Which plant tissue mainly transports water from roots to leaves?

A. phloem
B. xylem
C. epidermis only
D. pollen

Question 11 1 mark Medium Biology MCQ

11. Pollination is the transfer of pollen from the anther to the:

A. root
B. stigma
C. ovary wall only
D. leaf blade

Question 12 1 mark Hard Chemistry MCQ

12. A catalyst generally:

A. is consumed completely in every reaction
B. increases reaction rate without being consumed overall
C. changes the identity of every product
D. stops all collisions

Section — 2-mark questions

Question 13 2 marks Easy Chemistry Short Answer

13. What is a molecule? Give one example.

Question 14 2 marks Easy Biology Short Answer

14. Name the gas plants absorb during photosynthesis.

Question 15 2 marks Easy Physics Short Answer

15. Define acceleration and write its formula.

Question 16 2 marks Easy Chemistry Short Answer

16. What are isotopes? Give one example.

Question 17 2 marks Easy Biology Short Answer

17. State two functions of stomata in leaves.

Question 18 2 marks Easy Biology Short Answer

18. Distinguish a producer from a consumer in a food chain.

Question 19 2 marks Medium Physics Short Answer

19. A 6 Ω resistor carries a current of 2 A. Calculate the potential difference across it.

Question 20 2 marks Medium Chemistry Short Answer

20. Balance the equation: $Mg + O_2 \rightarrow MgO$.

Question 21 2 marks Medium Chemistry Short Answer

21. A sample has a mass of 20 g and a volume of 5 cm^3. Calculate its density.

Section — 5-mark questions

Question 22 5 marks Medium Physics Long Answer

22. Describe the energy changes that occur when a ball is thrown vertically upward and falls back down, ignoring air resistance.

Question 23 5 marks Medium Biology Long Answer

23. Describe the path of blood through the heart, lungs, and body in one complete circulation.

Question 24 5 marks Hard Biology Long Answer

24. Explain how the structure of alveoli supports efficient gas exchange in human lungs.

Question 25 5 marks Hard Physics Long Answer

25. A 4 kg trolley accelerates at 5 m/s² while friction opposing motion is 6 N. Calculate the net force and the applied force, showing your reasoning.

Total: 50 marks